

A Data-Driven Approach to Customer Churn Prediction and Retention Optimization Using Machine Learning and A/B Testing

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Abstract—This paper addresses the problem of customer churn in E-commerce using a data-driven approach. We develop machine learning models, including Random Forest and XGBoost, to predict churn probability and identify high-risk customers. Furthermore, we design an A/B testing framework to evaluate the causal impact of retention strategies. The results demonstrate statistically significant improvements in key business metrics such as Average Revenue Per User (ARPU). Based on both statistical and business significance, we provide actionable recommendations for system-wide deployment.

Index Terms—Churn Prediction, Customer Lifetime Value, A/B Testing, Machine Learning, Marketing Analytics

I. INTRODUCTION

Customer churn is one of the most critical challenges in E-commerce. Retaining existing customers is often more cost-effective than acquiring new ones. This study aims to develop a predictive and experimental framework to reduce churn and optimize retention strategies.

II. EXPLORATORY DATA ANALYSIS

Key findings from the dataset include:

- Revenue distribution is highly right-skewed, requiring log transformation.
- Customer behavior follows distinct RFM (Recency, Frequency, Monetary) segments.
- Significant class imbalance exists in churn labels.

III. MODELING

We apply machine learning models to estimate the probability of churn:

$$P(\text{churn} | X)$$

A. Feature Engineering

Key features include:

- Recency, Frequency, Monetary (RFM)
- Behavioral activity (clicks, sessions)
- Temporal trends (activity slope)

B. Evaluation Metrics

- Precision-Recall AUC (PR-AUC)
- F2-score (for imbalanced classification)

IV. EXPERIMENTAL DESIGN

To evaluate real-world impact, we design an A/B testing framework.

A. Hypothesis

- H_0 : No difference between control and treatment
- H_1 : Treatment improves retention

B. Metrics

- Primary Metric: Average Revenue Per User (ARPU)
- Guardrail Metrics: Customer complaints, system latency

C. Statistical Setup

- Significance level: $\alpha = 0.05$
- Statistical power: $1 - \beta = 0.8$

V. RESULTS

Metric	Control (A)	Treatment (B)	Lift
Sample Size	15000	15000	-
ARPU	200	215	+7.5%
p-value	-	-	< 0.01

TABLE I
A/B TESTING RESULTS

The treatment group shows statistically significant improvement in ARPU, indicating positive impact of the retention strategy.

VI. CONCLUSION

This study demonstrates that combining machine learning with A/B testing provides a robust framework for churn reduction. The results support deployment of the proposed system, subject to cost-benefit validation.

REFERENCES

- [1] F. Provost and T. Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
- [2] K. P. Murphy, *Machine Learning: A Probabilistic Perspective*, MIT Press, 2012.
- [3] R. Kohavi et al., *Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing*, Cambridge University Press, 2020.
- [4] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [5] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. KDD*, 2016.